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Note on Adjusted Present Value

A standard procedure for evaluating a project is to forecast incremental after-tax cash flows, excluding payments to capital providers, and discount them at a weighted average cost of capital (WACC) to obtain an estimate of the project's net present value (NPV). This note presents an alternative procedure for estimating NPV, namely the adjusted present value (APV) technique. APV is *not* a substitute for NPV as a capital budgeting decision criterion. Rather, it is an approach to computing NPV that is sometimes simpler, more accurate, or more informative than using the WACC. APV is sometimes referred to as "valuation in parts" or "valuation by components."[1]

Many practitioners have been trained to use the WACC, which has considerable intuitive appeal. Accordingly, this note begins by reconciling a valuation based on the WACC with one based on APV. It then identifies conditions necessary for the WACC approach to give reasonable estimates. When those conditions do not hold, APV will often yield a better estimate. Sometimes a WACC-based approach might work, but APV is either simpler or more compatible with the way managers think about executing a corporate financial program. The note concludes with some practical observations about using APV in common corporate valuation problems.

Comparing APV and WACC

Consider a simple, one-year project with the following characteristics. At the beginning of the year, a cash outlay of $500 is required, none of which is depreciable or otherwise deductible for tax purposes. One year later the project produces earnings before interest and taxes (EBIT) of either $1,200 or $550, with equal probability. Hence, the expected EBIT is $875. Assume the project's net salvage value at the end of the year is zero and the corporate tax rate is 34% (ignore personal taxes for the moment). Let the market return on a one-year risk-free bond be 6%.

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[1]APV was first developed and presented by Stewart C. Myers in "Interactions of Corporate Financing and Investment Decisions—Implications for Capital Budgeting," *Journal of Finance*, vol. 29, pp. 1-25, March 1974. For more on APV and other alternatives to the WACC, see Chapter 19, pp. 457-76 in Richard A. Brealey and Stewart C. Myers, *Principles of Corporate Finance*, 4th ed., McGraw-Hill, Inc., New York, 1991.

Professor Timothy A. Luehrman prepared this note as the basis for class discussion rather than to illustrate either effective or ineffective handling of an administrative situation.

The APV Approach

APV separates project value into one component associated with the unlevered project, and another associated with its financing program. Each component is evaluated separately. The APV is simply the sum of the two components.

$$\text{APV} = \boxed{\begin{array}{c}\text{Present value of the}\\ \text{all-equity-financed project}\end{array}} + \boxed{\begin{array}{c}\text{Present value of the side effects}\\ \text{associated with the financing program}\end{array}}$$

The after-tax cash flows for this project are simply -$500 at the beginning of the year and $875 x (1- 0.34) = $578 at the end of the year. The present value of the unlevered project depends on the opportunity cost of committing funds to it. Suppose the market return on a one-year investment of comparable risk is 12%. Then the value of the project equals = ($578/1.12) = $516, which is greater than the required investment of $500.

The project is even more valuable than this calculation shows, however, because of its debt capacity. Note that the project owners could borrow $349 at the risk-free rate of 6%. Debt up to this amount is risk-free because the worst possible outcome is EBIT = $550. Interest on the debt would be $349 x 0.06 = $21, so worst-case earnings before taxes would be $550 - $21 = $529. Taxes at 34% come to $180. This leaves worst-case profit after interest and taxes of $529 - $180 = $349, enough to pay off the debt at the end of the year.

Suppose the project owners decide to undertake the project and finance it with $349 of debt and $151 of equity.[2] The deductibility of interest creates a tax shield equal to the expected interest payment times the tax rate ($21 x 0.34 = $7). Because the debt is risk-free, the cost of financial distress is zero. Therefore, the entire $7 of value from the interest tax shield is realized by the project owners at the end of the year. The present value of the shield is $7/1.06 = $6.6 ≈ $7.[3]

The APV of the project is simply the value of the project if financed entirely with equity plus the value of the tax shield created by debt financing: APV = $516 + $7 = $523. Therefore, NPV = $523 - $500 = $23.

The WACC Approach

Now compute the WACC for the same project and capital structure. Recall that:

$$\text{WACC} = (D/V)k_d(1\text{-}t) + (E/V)k_e,$$

where D, E, and V denote the market values of debt, equity, and the whole project, respectively. Debt comprises 70% of the financing and equity, 30%.[4] The tax rate, t, is 34% and the cost of debt, k_d, is 6%. The cost of levered equity, k_e, is the risk-free rate plus a premium for the business

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[2]This is not to say that $349 of debt represents the *optimal* capital structure. It might or might not, depending on the cost of financial distress at higher levels of debt and, in a richer setting, certain other considerations.

[3]The tax shield is discounted here at the risk-free rate because it is riskless. It equals $7 at the end of the year no matter which EBIT outcome occurs. For most companies, debt is not riskless and neither are interest tax shields, which are commonly discounted at the cost of debt under the hypothesis that they are comparably risky.

[4]Strictly speaking, WACC requires the debt ratio to be expressed as a fraction of (levered) firm value, not project cost, so that D+E=V. In this case, the distinction does not materially affect the estimated NPV.

and financial risk borne by the equity investors. For the all-equity capital structure considered above the risk premium was 12% - 6% = 6%, reflecting only business risk. In the levered firm, the same amount of business risk is borne by much less equity, and a correspondingly higher risk premium is necessary. The risk premium for the levered equity is (V/E) x (premium for unlevered equity) = (1/0.30) x 6% = 20%.[5] Adding this premium to the risk-free rate gives the cost of levered equity: 6% + 20% = 26%. Putting all the pieces together, WACC = 0.70(6%) x (1- 0.34) + 0.30(26%) = 10.6%.

Discounting the project's unlevered cash flow of $578 by the WACC of 10.6% gives a project value of $523. Subtracting the initial investment of $500 gives an NPV of $23. This is the same answer obtained using the APV approach.

Advantages of APV

It is, of course, no accident that these two methodologies give the same result in such a simple setting. The difference between them is solely in their treatment of the value of leverage. Note that for an unlevered firm or project, the two use exactly the same cash flows and discount rates. For a levered firm, APV treats tax shields and other potential side effects of leverage as *separate cash flows,* which are discounted separately and then added to the all-equity-financed value of the project. In contrast, the WACC approach adjusts the *discount factor* rather than the cash flows. It applies a somewhat lower, "tax-adjusted" discount rate to all the unlevered project cash flows. A firm with an optimal capital structure should have a WACC that is a bit lower than its unlevered cost of equity.

If APV and WACC give the same answer, why prefer one over the other? There are three primary reasons. First, the two methods do *not* always give the same answer. The WACC approach yields poor estimates when certain assumptions fail to hold. APV does not rely on these assumptions, and remains valid when they do not hold. Second, practitioners routinely miscalculate WACC, sometimes producing large, unnecessary errors in their estimates of value. APV is less prone to such miscalculations. Finally, APV provides a potentially useful unbundling of components of value.

Before describing APV's advantages more fully below, we should note that WACC has advantages of its own. These do not add up to a compelling case for routinely preferring WACC over APV, but they are advantages nonetheless. First, WACC is widely used, so a WACC-based valuation may be easily communicated throughout a firm's internal capital budgeting apparatus. This advantage is potentially quite important, but it is eroding as more people become trained to use APV. Second, WACC is a natural way to evaluate a company that strictly maintains its debt at a constant fraction of firm value. In reality, even though many managers announce this policy, few appear to practice it. [Note, for example, that it implies issuing equity or repurchasing debt following a drop in stock price.] Finally, even though computing the WACC itself can be complicated, only one discounting calculation is required by this approach, whereas APV may require several. This was perhaps a significant efficiency in the era of slide rules and early calculators, but it is surely almost irrelevant now.

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[5]The relationship between risk premia for levered and unlevered equity is the familiar formula used to adjust a stock's ß for changes in leverage. The version used here does not hold exactly when debt is risky (here the debt is riskless), and it may not hold in the presence of personal taxes. For more on these circumstances, see Brealey and Myers, *op.cit.*, Chapter 9, pp. 181-212, and Chapter 18, pp. 426-434.

Violations of Assumptions

APV and WACC both require value additivity. WACC requires several additional assumptions that, unfortunately, often fail to hold. First, using WACC for a project requires that the project look just like the firm undertaking it (or just like some "twin" firm), with respect to both risk and capital structure. Many managers and analysts understand this requirement, but the WACC formula naturally leads them to exploit firm-level data, such as debt and equity ratios from consolidated balance sheets, and the firm's own beta for computing a cost of equity. But in reality, large investment projects seldom look just like the firm. Indeed, major investment programs are often intended to change the firm—to revitalize it, or to effect a change in strategy. Using WACC with firm-level inputs may yield a poor estimate of the value of such projects.

Second, in a similar vein, WACC will generally miss or distort valuable, project-specific financing or tax opportunities. Common examples are tax breaks and subsidized financing associated with certain activities (e.g., exporting), or investments in certain locations or types of facilities (e.g. tax-exempt bonds). Such features of a project make it inappropriate to use the corporate tax rate, debt ratio, and/or cost of debt in the WACC formula without additional adjustments.

Third, for WACC to correctly reflect the value of interest tax shields, it must be the case that the corporation's tax deduction for interest expense equals the cost of debt times the market value of the debt. There are many corporate debt securities for which this has not been the case at one time or another, including zero-coupon bonds, other original-issue-discount bonds, and some convertible bonds, junk bonds, and tax exempt bonds. To the extent that the available interest deduction differs from $k_d D$, the WACC will misestimate the value of interest tax shields.

Finally, WACC works best for cash flows that approximate a steady-state perpetuity; that is, for projects in which risk, leverage, and growth are basically constant for a long period. For other patterns of cash flows, WACC still can be made to work under the very restrictive assumption that debt comprises a constant fraction of firm value. This assumption is violated by projects or firms that move through cycles or stages, such as a start-up or product launch; by large project financings; and by unusual events or transactions in a company's history, such as a restructuring or leveraged buyout. Theoretically, the WACC can be recomputed throughout the project's life as required by changing project characteristics. But such repeated recalculations are at least as difficult and no more accurate than a similar exercise using APV.

Miscalculations by Practitioners

Practitioners sometimes miscalculate WACC by treating its separate components as independent or exogenous. In fact, although the debt and equity ratios may be chosen by managers, the corresponding costs of debt and equity are determined by prevailing capital market conditions, *given* the chosen capital structure. Consequently, it is improper to adjust the capital structure in the WACC formula without making corresponding adjustments to the costs of debt and equity.

To see the importance of this requirement, return to the numerical example presented above. In that example, the cost of capital for an all-equity capital structure was 12%. The WACC for this capital structure is, similarly, 12%: WACC = 0.00(6%)(1 - 0.34) + 1.00(12%) = 12%. Now suppose a manager, knowing the firm could borrow at the risk-free rate, decided to lower its cost of capital by levering the project up to 70% debt. He or she might recompute the cost of capital as: WACC = 0.70(6%)(1 - 0.34) + 0.30(12%) = 6.4%. This, of course, is incorrect because the cost of equity cannot be 12% for both the levered and unlevered capital structures. As shown above, the cost of equity for this project with 70% debt is 26%, not 12%, and the WACC is 10.6%, not 6.4%. The computational error of 420 basis points is huge compared to the correct WACC.

Further, the true difference between the WACC for 0% debt and 70% debt is only 1.4%, not 5.6%. Finally, using the wrong WACC gives an NPV of ($578/1.064) - $500 = $43, rather than $26, an error of 65% in the estimate of NPV.

The example just presented is a dramatic one, but subtle versions of the same error are commonplace. They all have the same source, a failure to observe the interdependence of the individual elements of the WACC. Whatever its size, such an error is unnecessary. It does not reflect poor data, but poor execution. It can be avoided by taking care or by using the APV approach.

Estimating the cost of debt, especially at high debt ratios, presents another trap in using the WACC formula. A common, but misguided, practice is to use the coupon rate on the riskiest debt in the capital structure as the cost of debt. What the formula really requires is k_d, the expected return on the debt, which may be quite different from the coupon rate. This is obviously the case for debt that does not trade at par. Yield-to-maturity might be a better estimate for such instruments, but this also is imperfect. Yields are computed based on *promised* payments to lenders, not *expected* payments. Investors in risky debt, junk bonds for example, might not be paid all they were promised if the issuer defaults. Assuming they understand this and price the bonds accordingly, their expected return is less than the yield-to-maturity as typically calculated. Accordingly, yield is a biased estimate of the cost of risky debt.

Finally, high leverage is frequently accompanied by complexity, in the number of layers of debt or the features of individual issues or both. In such cases, the expected return (even if perfectly estimated) on the riskiest issue does not equal the firm's cost of debt. A better approximation would be the expected return on a "strip", or value-weighted portfolio, of each issue. Unfortunately, this can be very difficult to compute.

Unbundling Components of Value

APV preserves information about components of value that is easily lost in the aggregation that a WACC-based approach employs. In the simple example presented above, both APV and WACC give an NPV of $23, but APV also shows that $7 of the $23 (30%) comes from interest tax shields. This additional information may or may not be managerially useful in such a simple project, but it is neither harmful nor expensive. In a complicated project it may be essential as a way to better understand and manage the project. This is especially the case for projects that involve predictable changes over time, and that are expected to be financed differently at different points in time.

In a similar way, the capital budgeting and investment process itself is comprised of a sequence of tasks, only one of which is deciding whether or not to invest. Schematically, a typical project goes through several stages:

Identification → Development → Decision → Implementation → Management

At the point of decision, WACC may be outperformed by APV. But even if it is not—that is, even if the necessary assumptions hold and WACC is not miscalculated—the unbundling of components of value provides useful information at other points in the process. For some projects, it may be helpful to extend the disaggregation of value even further and modify the simple definition of APV presented above. Distinguishing between value associated with the project on the one hand and its financing on the other is a logical way to disaggregate value, but it is not the only one. Different decisions that must be made about a project's fate may suggest other useful separations as well. For example:

"Modified" APV	=	Base case value: All-equity financed, assets-in-place only, excluding growth options, excluding all subsidies	+	Present value of growth options	+	Present value of subsidies (non-financing only)	+	Present value of all financing side effects

In theory, cash flows associated with each of these pieces of value may be estimated over the life of the project, and discounted separately at an opportunity cost that reflects their risk characteristics.[6] Each piece provides input for different, albeit related, decisions. One helps in designing a financing program and negotiating with investors; another helps prepare for discussions with regulators and local government authorities; another is employed in the strategic planning and corporate development processes. For a given project, the most appropriate division of value for a modified APV depends on the different categories of value actually present, the quality of available data, and the particular decisions to be made. Used this way, APV supplies managers with more than simply an answer to the question, "What is the NPV?"

Using APV

APV always works when a WACC approach works and it sometimes works when WACC does not. Nevertheless, APV remains a "discounted cash flow" (DCF) method of estimating value. It assumes that the riskiness of future cash flows is such that it can be properly reflected in a stable discount rate. When DCF is inappropriate as a methodology, so is APV (and WACC). The most important such instance is when the project being evaluated is essentially an option. Buying a building is fundamentally different from buying an option on the same building. DCF is a good way to evaluate the building, but it is poorly suited to evaluating an option on the building. APV does not escape this fundamental limitation of DCF valuation techniques.

APV requires users to become comfortable valuing the "side effects" of a financing program in isolation. These include not only interest tax shields, but also loan subsidies, loan guarantees, issue costs for new securities, the cost of financial distress, and so forth. Even the most common and straight-forward of these, interest tax shields, is actually difficult to evaluate.[7] The value of subsidies and guarantees depends on the terms and conditions under which they are available and may be complex indeed. WACC-based analyses typically ignore such complexities altogether. APV at least provides a "box" for the analyst to put them in, though in practice they are often omitted from APV computations as well.

Like almost any analytical tool, APV can be misapplied. Perhaps the most common problem is an overzealous division of value into ever-smaller subparts. Analysts may be tempted to value a firm product by product, or market by market, or to discount inflows separately from outflows, and so forth. The possibilities are endless. But the quality of the data may diminish quickly with successive divisions. Certainly opportunities to commit fundamental errors increase as the exercise gets more intricate. It may become easier, for example, to mistake income for cash flow, or misallocate or double count some items.

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[6]The present value of growth options is an exception. This component should be evaluated with option pricing techniques rather than the standard discounted cash flow technique. Option pricing is beyond the scope of this note, but see, for example, Brealey and Myers, *op.cit.*, Chapters 20 and 21, pp. 483-534.

[7]Brealey and Myers, *op.cit.*, Chapter 18, pp. 426-34 describe the difficulties of valuing interest tax shields.

An additional hazard of extensive partitioning is that, in treating each group of cash flows separately, an analyst might fail to check whether the pieces form a coherent whole. Suppose, for example, all the assets of a firm together have an unlevered beta (or "asset beta") of about 0.7, based on fairly reliable data. Someone might want to discount the constituent cash flows separately, believing that different cash flows had different unlevered betas. Note, though, that however these "sub-betas" are estimated, a value-weighted average of them must be roughly equal to 0.7. Without some information or experienced intuition about differences in risk, an analyst might just as well *examine* the disaggregated cash flows (and note patterns over time, etc.), but perform the actual discounting at the level of the all-equity-financed firm or project.

In summary, APV has some pitfalls, but most arise in only fancy formulations. In its most basic form, which separates the value of the project from the value of its financing program, APV is easy to use. Ease of use, together with wider applicability and less scope for misuse than WACC, makes it likely that APV eventually will become the DCF methodology of choice among practitioners.